

1. Core Java (Advanced)

Object-Oriented Design

- SOLID principles
 - Composition vs Inheritance
 - Design by Contract
 - Immutable objects
 - Defensive copying
 - Value objects
 - Records
 - Sealed classes
 - Enums
-

Generics

Advanced generics including

- Type erasure
- Wildcards
- PECS principle
- Generic methods
- Generic classes
- Bounded types
- Recursive generics
- Reflection with generics

Example

```
public <T extends Number> void process(List<? extends T> list)
```

Collections Framework

You should know internals.

- ArrayList
- LinkedList

- HashMap
- LinkedHashMap
- TreeMap
- HashSet
- TreeSet
- ConcurrentHashMap
- PriorityQueue
- CopyOnWriteArrayList

Internals

- Hashing
 - Load factor
 - Rehashing
 - Treeification
 - Collision handling
 - Red-black tree
 - Iterator implementation
 - Fail-fast
 - Fail-safe
-

String Internals

- String Pool
 - StringBuilder
 - StringBuffer
 - UTF-8
 - UTF-16
 - Compact Strings (Java 9)
 - Interning
-

Exception Handling

- Checked vs Unchecked
- Custom exceptions
- Suppressed exceptions
- try-with-resources
- Exception hierarchy
- Best practices

2. JVM Internals

Very important.

Class Loading

- Bootstrap ClassLoader
- Platform ClassLoader
- Application ClassLoader
- Parent delegation
- Custom ClassLoader

JVM Memory

Heap

Young Generation

Eden

Survivor

Old Generation

Metaspace

Thread Stack

Native Memory

Know

- Object allocation
 - Escape analysis
 - Stack allocation
 - Object header
 - Pointer compression
-

Garbage Collection

Must know

Serial GC

Parallel GC

CMS

G1GC

ZGC

Shenandoah

Understand

- Minor GC
 - Major GC
 - Full GC
 - STW
 - GC Roots
 - Reachability analysis
-

JVM Performance

- JIT compiler
 - C1
 - C2
 - Tiered compilation
 - HotSpot
 - Inlining
 - Escape analysis
-

3. Multithreading & Concurrency

Huge topic.

Thread Basics

- Thread lifecycle
 - Runnable
 - Callable
 - Future
 - FutureTask
-

Synchronization

- synchronized
- volatile
- Atomic classes
- Locks

ReentrantLock

ReadWriteLock

StampedLock

Java Memory Model

Must understand

- Happens-before
 - Visibility
 - Atomicity
 - Reordering
 - Memory barriers
-

Concurrent Collections

- ConcurrentHashMap
- BlockingQueue
- ConcurrentLinkedQueue
- CopyOnWriteArrayList

Executors

- FixedThreadPool
 - CachedThreadPool
 - ScheduledExecutor
 - ForkJoinPool
-

CompletableFuture

Must know deeply.

- Async pipeline
 - thenApply
 - thenCompose
 - allOf
 - anyOf
 - exceptionally
-

Virtual Threads (Java 21)

- Project Loom
- Structured Concurrency
- Scoped Values

Very hot topic.

4. Functional Programming

- Lambda
- Functional Interface
- Method Reference
- Streams API
- Parallel Streams
- Optional

Know stream internals

- Lazy evaluation
 - Splitterator
 - Collector
-

5. Reflection

- Class API
 - Method API
 - Constructor API
 - Field API
 - Dynamic invocation
 - Annotation processing
-

6. Annotations

- Custom annotations
 - Retention policy
 - Annotation processors
 - Runtime annotations
-

7. Serialization

- Java Serialization
 - JSON serialization
 - Externalizable
 - Serializable
 - Versioning
-

8. Java I/O

Old IO

InputStream

OutputStream

Reader

Writer

NIO

Path

Files

Buffer

Channel

Selector

9. Networking

- TCP
 - UDP
 - Socket Programming
 - HTTP
 - HTTPS
 - WebSocket
-

10. Security

- SSL
- TLS
- JWT
- OAuth2
- OpenID Connect
- Encryption

- Hashing
 - KeyStore
 - TrustStore
-

11. Spring Framework

Must know.

Core

- IoC
 - Dependency Injection
 - Bean lifecycle
 - Bean scopes
-

Spring Boot

- Auto Configuration
 - Starter
 - Profiles
 - Properties
 - Configuration
-

Spring MVC

- DispatcherServlet
 - Controllers
 - Filters
 - Interceptors
-

Spring Data

- JPA
 - Repository
 - Specifications
-

Spring Security

- Authentication
 - Authorization
 - JWT
 - OAuth
-

Spring AOP

- Aspect
 - Advice
 - Join Point
 - Proxy
-

12. Database

SQL

Must know

- Indexes
 - Execution plan
 - Joins
 - Transactions
 - Isolation levels
 - Deadlocks
 - Locks
-

JPA/Hibernate

- Entity Lifecycle
- Lazy Loading
- Eager Loading
- Dirty Checking
- Cache
- N+1 problem
- JPQL
- Criteria API

13. Microservices

- Service discovery
- API Gateway
- Circuit Breaker
- Retry
- Bulkhead
- Saga
- Event-driven architecture
- CQRS
- Outbox Pattern

14. Distributed Systems

- CAP theorem
- BASE
- Consistency
- Consensus
- Leader Election
- Distributed Lock
- Idempotency

15. Messaging

- Kafka
- RabbitMQ
- ActiveMQ
- JMS

Need to know

- Partition
- Consumer Group
- Offset

- Ordering
 - Exactly Once
 - At Least Once
-

16. Caching

- Redis
- Ehcache
- Caffeine

Concepts

- Cache Aside
 - Read Through
 - Write Through
 - Write Back
-

17. Design Patterns

Creational

- Singleton
- Builder
- Factory
- Prototype

Structural

- Adapter
- Bridge
- Composite
- Decorator
- Facade
- Proxy

Behavioral

- Observer
- Strategy

- Command
 - Chain of Responsibility
 - Template
 - Visitor
 - State
-

18. System Design

Must know.

- Load Balancer
 - Reverse Proxy
 - API Gateway
 - CDN
 - Sharding
 - Replication
 - Partitioning
 - Scaling
 - Consistent Hashing
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19. Performance Tuning

- JVM tuning
- GC tuning
- Thread dump
- Heap dump
- Profiling
- Memory leak detection

Tools

- VisualVM
 - JProfiler
 - Java Flight Recorder (JFR)
 - Java Mission Control (JMC)
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20. Testing

- JUnit
 - Mockito
 - Integration testing
 - TestContainers
 - WireMock
 - Performance testing
-

21. Build Tools

- Maven
 - Gradle
 - Dependency Management
 - Multi-module projects
 - Plugins
-

22. DevOps

- Docker
 - Kubernetes
 - CI/CD
 - GitHub Actions
 - Jenkins
 - Helm
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23. Observability

- Logging
- Metrics
- Tracing

Tools

- ELK Stack
 - Prometheus
 - Grafana
 - OpenTelemetry
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24. Cloud

- AWS
- Azure
- GCP

Know

- Containers
 - Serverless
 - Object storage
 - IAM
 - Networking
-

25. AI Integration (Growing Importance)

Since you've recently been working with Retrieval-Augmented Generation (RAG) and code migration, modern Java developers should also understand:

- LLM fundamentals
 - Embeddings
 - Vector databases (Qdrant, OpenSearch vector search, pgvector)
 - RAG architecture
 - AI agent frameworks (e.g., LangChain4j, Spring AI)
 - Model Context Protocol (MCP)
 - Prompt engineering
 - Tool calling and function calling
 - Model evaluation and observability
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26. Enterprise Integration

- REST API design
 - gRPC
 - GraphQL
 - OpenAPI / Swagger
 - XML & JSON processing
 - Apache Camel
 - Batch processing
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27. Advanced Language Features (Java 8–21)

A senior developer should be comfortable with features introduced across recent Java releases:

- Lambdas and Streams
- Default and static interface methods
- `var`
- Records
- Sealed classes
- Pattern matching for `instanceof`
- Pattern matching for `switch`
- Text blocks
- Switch expressions
- Helpful `NullPointerException` messages
- Virtual threads
- Structured concurrency (preview)
- Scoped values (preview)
- Foreign Function & Memory API (when applicable)